Jaewook J. Suh

CONTACT INFORMATION

Department of Applied Mathematics & Statistics
Johns Hopkins University

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RESEARCH INTERESTS

AI- and Computer-Assisted Algorithm Analysis, Convex Optimization, Monotone Operator Theory, Adaptive First-Order Methods, Accelerated Algorithms, Optimization for Machine Learning

ACADEMIC APPOINTMENTS & AFFILIATIONS

Johns Hopkins University 2026 – Present
Lecturer, Department of Applied Mathematics & Statistics

Rice University 2024 – 2026
Postdoctoral Researcher
- Worked with Professor Shiqian Ma `[website]`

EDUCATION

Seoul National University 2024
Ph.D. in Mathematical Sciences
- Advisor: Professor Ernest K. Ryu `[website]`

Seoul National University 2018
B.S. in Mathematics and Physics
Teacher Certification in Mathematics and Physics

PUBLICATIONS

Published

1. **J. J. Suh**[*], T. Yoon[*], E. D. H. Nguyen, B. Ying, S. Ma. *Peppy: An* AI*-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms.* ICML Workshop: AI as a Tool for Mathematics, Computer Science, and Machine Learning, 2026. *Workshop version.* [*]Lead authors
2. R. Park, **J. J. Suh**, Y. Hong, E. K. Ryu. *Numerical Analysis of HiPPO-LegS ODE for Deep State Space Models.* Transactions on Machine Learning Research, 2026.
3. J. Park[*], **J. J. Suh**[*], B. Wang, A. Bhattacharya, S. Ma. *Adaptive Gradient Descent on Riemannian Manifolds and Its Applications to Gaussian Variational Inference.* ICLR, 2026. [*]lead authors
4. **J. J. Suh**, B. Ying, X. Jiang, E. D. H. Nguyen. *PEPFlow: A Python Library for the Workflow of Performance Estimation of Optimization Algorithms.* NeurIPS Workshop on GPU-accelerated and Scalable Optimization, 2025. `[Website]` `[GitHub]`
5. S. P. Boyd, T. Parshakova[*], E. K. Ryu, and **J. J. Suh**[*]. *Optimization Algorithm Design via Electric Circuits.* NeurIPS (spotlight, 2.1%), 2024. `[GitHub]` [*]Lead authors (alphabetical)
6. T. Yoon, J. Kim, **J. J. Suh**, and E. K. Ryu. *Optimal Acceleration for Minimax and Fixed-Point Problems is Not Unique.* ICML (spotlight, 3.5%), 2024.
7. **J. J. Suh**, J. Park, and E. K. Ryu. *Continuous-time Analysis of Anchor Acceleration.* NeurIPS, 2023.
8. **J. J. Suh**, G. Roh, and E. K. Ryu. *Continuous-time Analysis of Accelerated Gradient Methods via Conservation Laws in Dilated Coordinate Systems.* ICML (long presentation, top 2%), 2022.

Preprints

1. **J. J. Suh**, T. Yoon, E. D. H. Nguyen, B. Ying, S. Ma. *Peppy: An* AI*-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms.* arXiv preprint (full version), 2026.
2. T. Yoon[*], **J. J. Suh**[*], E. D. H. Nguyen, B. Ying, S. Ma. *Toward a Systematic Understanding and Interactive Search of Lyapunov-Style Proofs in Optimization.* 2026. [*]Lead authors

3. E. D. H. Nguyen, **J. J. Suh**, X. Jiang, S. Ma. *Exact worst-case convergence rates for Douglas–Rachford and Davis–Yin splitting methods.* 2025.
4. **J. J. Suh**, S. Ma. *An Adaptive and Parameter-Free Nesterov's Accelerated Gradient Method for Convex Optimization.* 2025.

Research Software & Tools

Peppy
- AI-assisted workflow for convergence analysis of optimization algorithms, built on PEPFlow

PEPFlow `[Website]` `[GitHub]` 2025 – Present
Co-developer
- Open-source Python library for computer-assisted analysis of optimization algorithms
- Provides infrastructure for interactively constructing convergence proofs via the PEP workflow
- Supports symbolic and numerical verification of worst-case convergence guarantees

Invited Talks & Seminars

Peppy: An AI-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms
- KAIST Graduate School of AI for Math Seminar — Jul 2026

Adaptive and Tuning-free First-Order Methods via Lyapunov Analysis
- Institute of Artificial Intelligence, University of Central Florida — May 2026
- Department of Combinatorics & Optimization, University of Waterloo — Feb 2026

Optimization Algorithm Design via Electric Circuits
- Quantitative Methods Seminar, Purdue University — Mar 2025
- KIAS Center for AI and Natural Sciences Seminar — Jan 2025

Teaching Experience

Lecturer, Johns Hopkins University
- Introduction to Convexity (EN.553.665) — Fall 2026

Substitute Lecturer, Rice University
- Iterative Methods for Systems of Equations and Unconstrained Optimization — Spring 2025

Instructor, Seoul National University
- Exercises in Introduction to Mathematical Analysis I — Spring 2020
- Exercises in Introduction to Mathematical Analysis II — Fall 2020
- Exercises in Calculus I — Spring 2019, Spring 2021
- Exercises in Calculus II — Fall 2018, Fall 2019, Spring 2022
  - Average student course evaluation score: 9.96/10

Professional Service

Conference Organization
- Minisymposium Organizer, SIAM Conference on Optimization (SIOPT), 2026
  - "AI- and Computer-Assisted Optimization"
- Session Organizer, INFORMS Optimization Society Conference (IOS) 2026
  - "Systematic Analysis of Optimization Algorithms Leveraging Computer-Assisted Proofs"

Reviewer
- ICML 2026 Workshop: AI as a Tool for Mathematics, Computer Science, and Machine Learning
- Journal of Machine Learning Research (JMLR)
- Journal of Computational and Graphical Statistics (JCGS)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- Mathematical Programming
- Journal of Optimization Theory and Applications

- Journal of Scientific Computing
- NeurIPS Workshop on GPU-accelerated and Scalable Optimization (ScaleOPT)

Conference Presentations

Peppy: An AI-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms

- ICML Workshop AI4Research (Poster) — Jul 2026

Adaptive Gradient Descent on Riemannian Manifolds and Its Applications to Gaussian Variational Inference

- International Conference on Learning Representations (Poster) — Apr 2026

PEPFlow: A Python Library for the Workflow of Performance Estimation of Optimization Algorithms

- SIAM Conference on Optimization — Jun 2026
- Cornell ORIE Young Researchers Workshop (Poster) — Oct 2025

An Adaptive and Parameter-Free Nesterov's Accelerated Gradient Method for Convex Optimization

- INFORMS Optimization Society Conference (IOS) — Mar 2026
- INFORMS Annual Meeting — Oct 2025
- International Conference on Continuous Optimization (ICCOPT) — Jul 2025

Optimization Algorithm Design via Electric Circuits

- AI-Owls meeting (Rice University) — Mar 2025
- Neural Information Processing Systems (Poster) — Dec 2024

Optimization Algorithm Design by Continuous-time Analysis

- KMS Spring Meeting — Apr 2024
- UNIST Workshop Optimization & Machine Learning — Aug 2023

Continuous-time Analysis of Anchor Acceleration

- Neural Information Processing Systems (Poster) — Dec 2023
- CJ CheilJedang Tech Talk — Jul 2023

Continuous-time Analysis of AGM via Conservation Laws in Dilated Coordinate Systems

- SIAM Conference on Optimization — May 2023
- Samsung AI Forum (Poster) — Nov 2022
- INFORMS Annual Meeting — Oct 2022
- International Conference on Machine Learning — Jul 2022

Awards & Honors

Postdoctoral Fellowship Program (Nurturing Next-Generation Researchers), granted by the National Research Foundation of Korea (NRF) — 2024

Google Travel Grant Award to attend ICML — 2022

Excellent Teaching Assistant Award — 2019

National Scholarship for Science and Engineering — 2013

Additional Teaching Assistant Experience

Teaching Assistant, Seoul National University

- Mathematical Foundations of Deep Neural Networks — Spring 2024
- Calculus for Life Science I — Fall 2023
- Differential Equations — Spring 2023
- Mathematical and Numerical Optimization — Fall 2021, Fall 2022
- Linear Algebra — Spring 2021
- Mathematics for Economics and Business — Spring 2020, Fall 2020
- Introduction to Mathematical Analysis I, II — Spring 2020, Fall 2020
- Engineering Mathematics I, II — Summer 2019, Fall 2019

- Calculus I, II — Fall 2018–Fall 2019, Spring 2021, Spring 2022

Teaching Practicum, Seoul National University Middle School (Physics) — May 2016
- Completed a one-month student teaching experience at a middle school
- Gained hands-on experience in classroom instruction and student engagement

Mentorship & Outreach

Undergraduate Research Mentorship
- Collaborated with undergraduate interns on research (published at ICML 2022, 2024)

Seoul National University College of Natural Science Camp (TA, Mathematics) — Aug 2019
Intercollegiate Education Outreach Club Activity — 2012–2014
- Visited various middle and high schools across the region to mentor students
- Provided studying tips on an online website through columns and Q&A
- Served as Club Officer (Fall 2013) and President (Spring 2014)